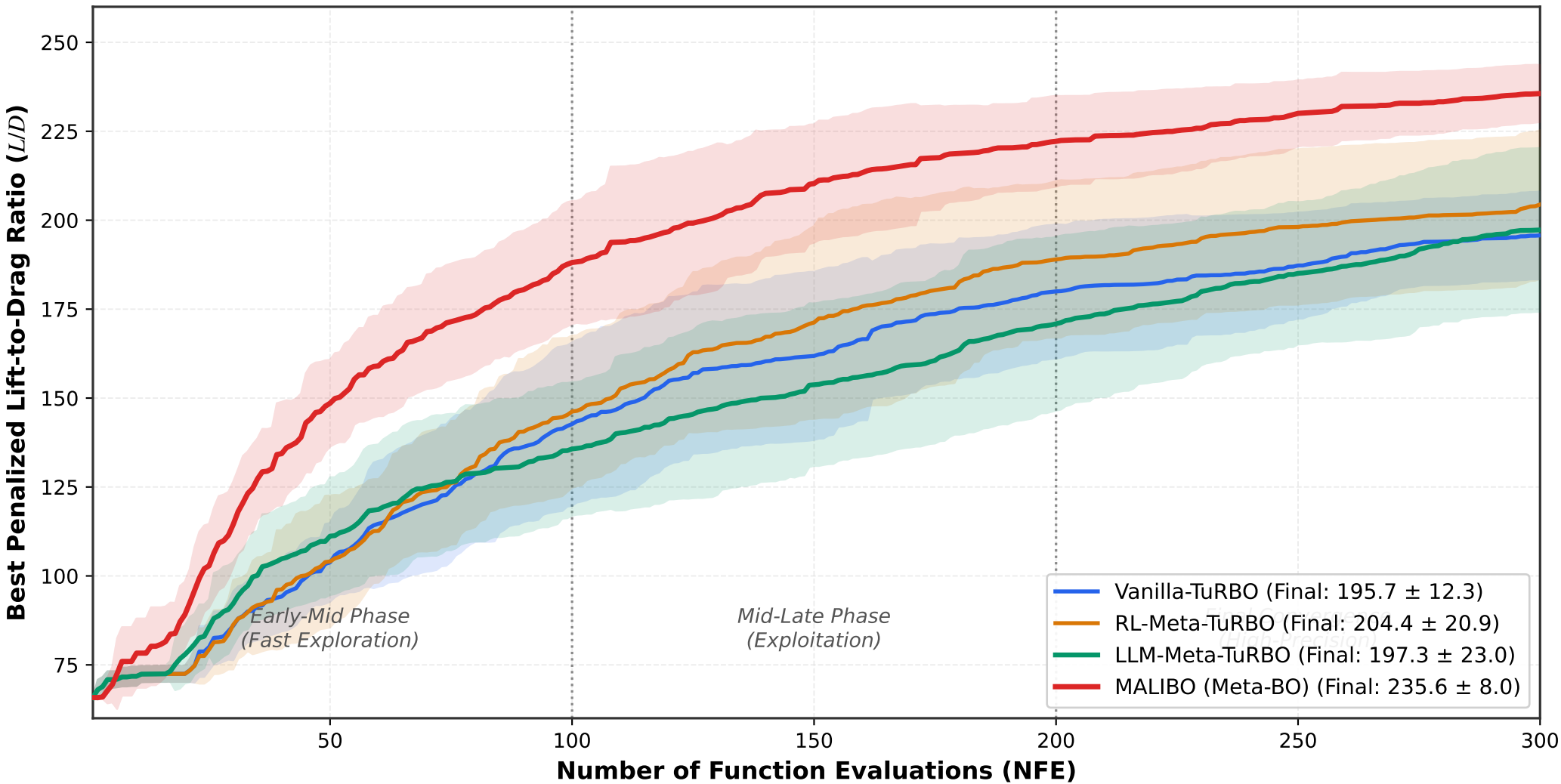

Mean Convergence Trajectories across 10 Seeds (Mean ± Std Shaded Band)
Flow Conditions: $Re = 3.5 \times 10^6, \alpha = 3.5^\circ$ | Initial Airfoil: NACA 0012
Best Penalized Lift-to-Drag Ratio (L/D)
Number of Function Evaluations (NFE)
250
225
200
175
150
125
100
75
50
100
150
200
250
300
Early-Mid Phase
(Fast Exploration)
Mid-Late Phase
(Exploitation)
Vanilla-TuRBO (Final: 195.7 ± 12.3)
RL-Meta-TuRBO (Final: 204.4 ± 20.9)
LLM-Meta-TuRBO (Final: 197.3 ± 23.0)
MALIBO (Meta-BO) (Final: 235.6 ± 8.0)